CARDANO BLOCKCHAIN | HACKATHON 2025

ExamGuard

Securing Exam Integrity

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The Problem

Exam integrity is under constant threat. Leaked papers, insider tampering, and a total absence of cryptographic proof have eroded public trust in examination systems worldwide.

Paper Leaks

Question papers sold or leaked before exam day, forcing cancellations.

Insider Threats

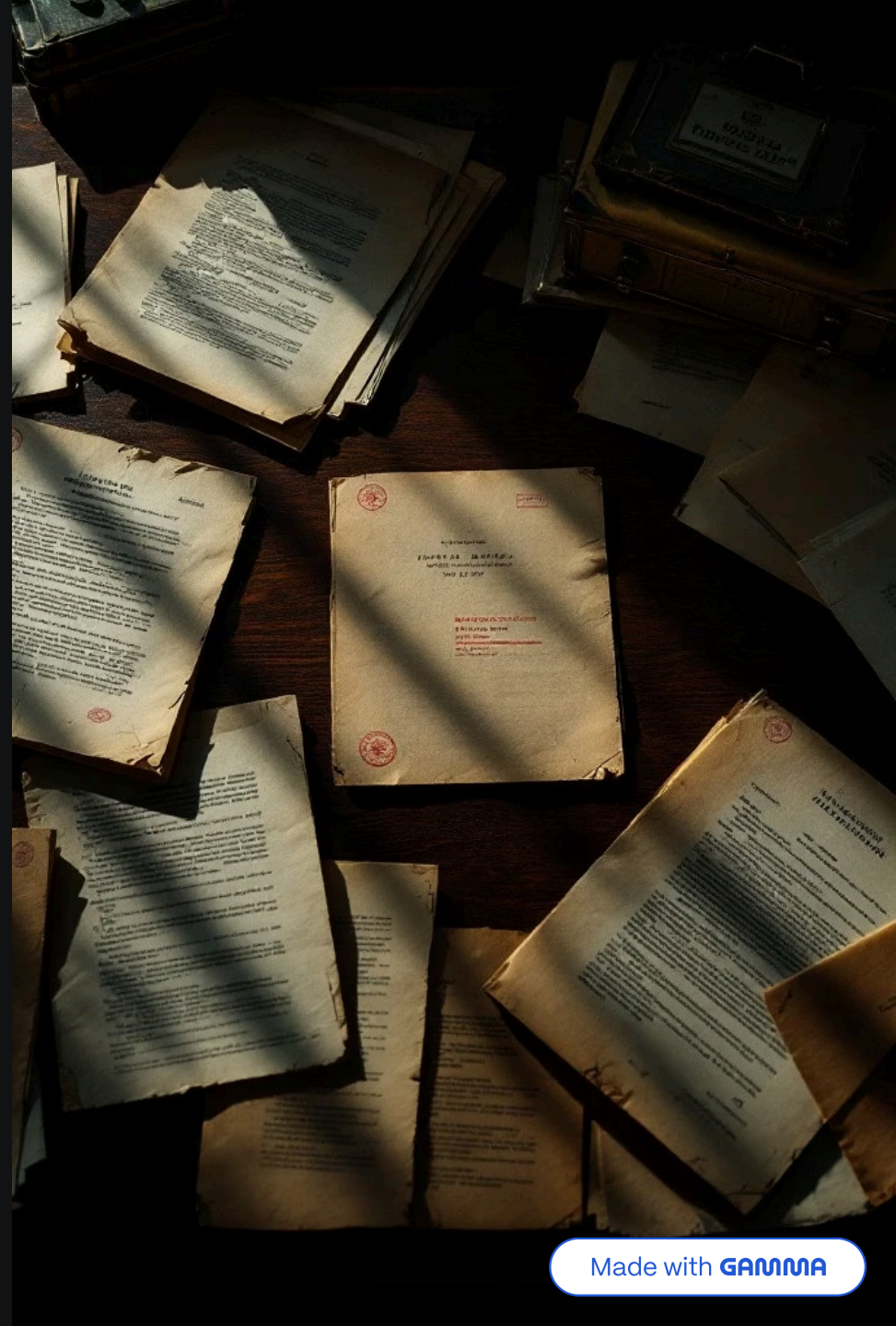
Privileged personnel can alter records without leaving any trace.

No Cryptographic Proof

No mechanism to verify that a paper was never modified after release.

Students Suffer

Cancelled exams delay careers and erode confidence in institutions.



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Why Traditional Systems Fail

Centralised databases are fast, but they fundamentally cannot guarantee immutability. Any administrator with sufficient privileges can silently edit, delete, or fabricate records — leaving no trace of tampering.

Capability	Traditional Database	Blockchain
Speed	✔️ Fast	✔️ Adequate
Immutability	❌ Editable	✔️ Immutable
Audit Trail	❌ None	✔️ Timestamped
Verification	❌ Closed	✔️ Public
Architecture	❌ Centralised	✔️ Decentralised

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Why Cardano?

Cardano was chosen for its academic rigour, energy efficiency, and predictable low-cost transactions — making it ideal for government-scale systems requiring public verifiability.

Energy Efficient

Proof-of-Stake consensus uses a fraction of the energy of traditional mining.

Low, Predictable Fees

Transaction costs are fractions of a cent — ideal for high-volume verification.

Peer-Reviewed Security

Built on academic research with formal verification of core protocols.

Developer Ecosystem

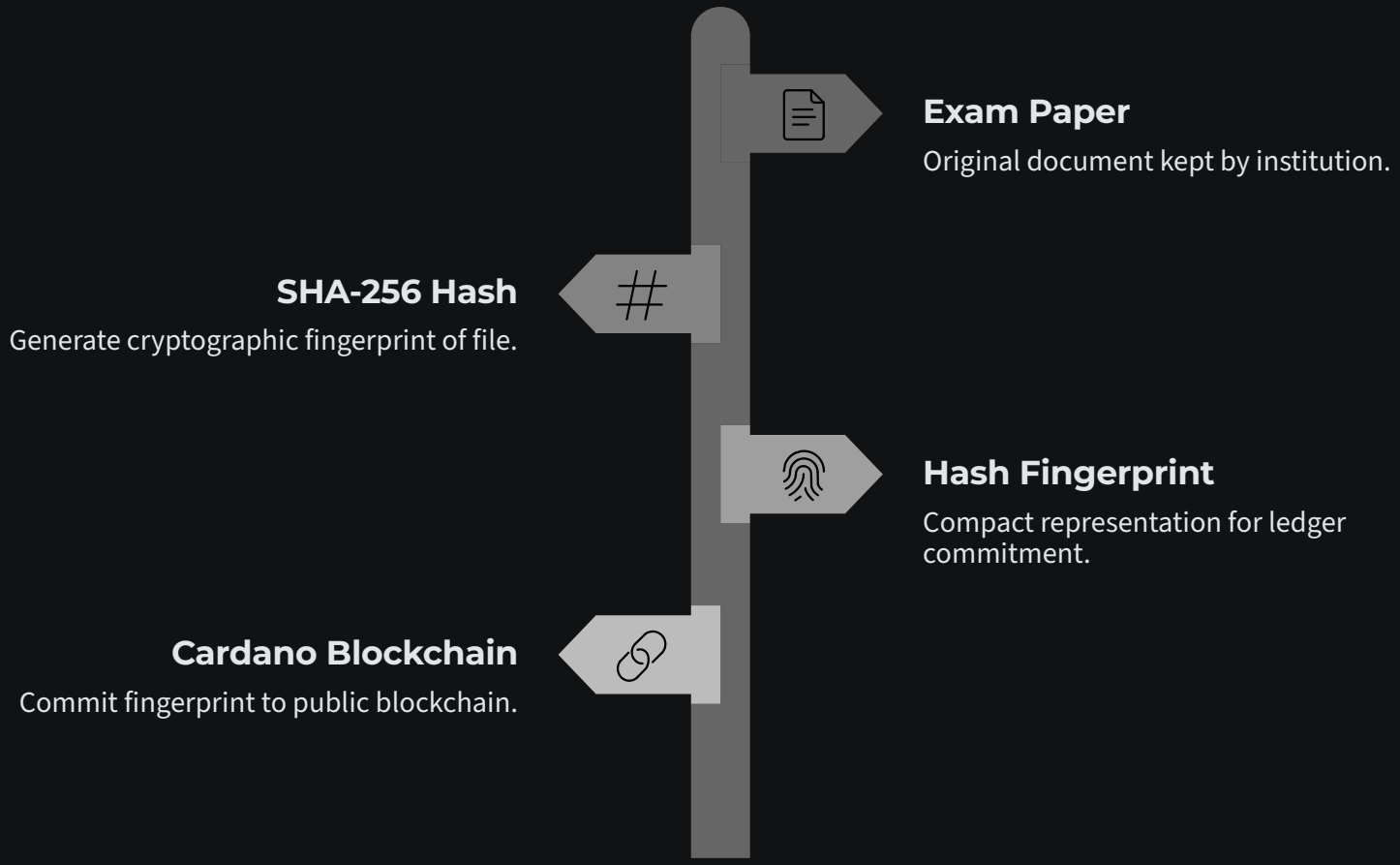
Mesh SDK and Blockfrost API provide robust, well-documented tooling.



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System Architecture

The exam paper never leaves the institution. Only its SHA-256 fingerprint is committed to the blockchain — creating a permanent, publicly verifiable anchor of trust.



Exam Paper

Original document kept by institution.

SHA-256 Hash

Generate cryptographic fingerprint of file.

Hash Fingerprint

Compact representation for ledger commitment.

Cardano Blockchain

Commit fingerprint to public blockchain.

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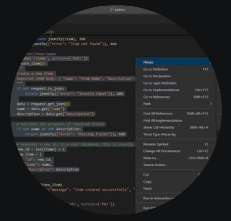
Tech Stack

A lean, modern stack designed for rapid prototyping and transparent verification. Each layer was chosen for reliability, accessibility, and developer experience.



Frontend

HTML, CSS, JavaScript



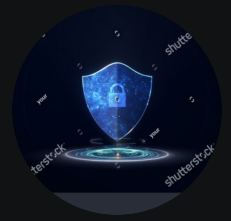
Backend

Node.js, Python



Blockchain

Cardano, Mesh SDK, Blockfrost



Cryptography

SHA-256 hashing



Deployment

GitHub, Vercel

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Project Structure

A modular repository where each file serves a distinct role in the pipeline — from hashing to on-chain submission to public verification.

1	**index.html** Verification interface for public users.
2	**pitch.html** Presentation deck for judges.
3	**hash_check.py** Generates SHA-256 fingerprints from PDFs.
4	**push_to_chain.js** Submits the hash as metadata via Mesh SDK.
5	**record.json** Stores transaction metadata locally.
6	**sample_exam.pdf** Demonstration document for the workflow.

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Live Workflow

From paper selection to on-chain confirmation — the entire verification pipeline completes in seconds.



Select Paper

Generate SHA-256

Store on Cardano

Confirm Tx

Verify Authenticity

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FAQ

Anticipated questions with concise, technically grounded answers.

Why Cardano?

Low fees, peer-reviewed security, and energy-efficient PoS ideal for public-sector adoption.

Why not a database?

Databases are mutable and centralised. Blockchain provides immutable, publicly verifiable proof.

One character changed?

The hash changes completely — verification fails instantly. That is the avalanche effect.

How scalable is this?

Each exam is one low-cost transaction. Cardano handles thousands per minute.

Why not store the PDF?

Cost and privacy. We store only the 64-character hash — the paper stays confidential.

Why SHA-256?

Collision-resistant, industry-standard, and deterministic. Any change produces a entirely different hash.

Production readiness?

Prototype is functional. Production needs key management, batch hashing, and audit dashboards.

Security considerations?

Private keys must be secured via hardware wallets. Metadata is public — no sensitive data is stored.

Can admins modify records?

No. Once committed, the transaction is permanent and tamper-evident on-chain.

Future improvements?

Multi-signature approval, automated batch verification, and institutional key rotation.

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Trust, Reimagined.

Immutable. Verifiable. Transparent.

ExamGuard transforms examination integrity from a promise into a cryptographic guarantee — built on Cardano, designed for institutions, and ready for the future of public trust.

Verify

Public, instant authentication of any exam paper.

Detect

Any tampering is immediately visible and provable.

Trust

Cryptographic proof replaces blind faith in institutions.



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